

Industrial wastewater source tracing: The initiative of SERS spectral signature aided by a one-dimensional convolutional neural network

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ABSTRACT

The spectral fingerprint is a significant concept in nontarget screening of environmental samples to direct identification efforts to relevant and important features. Surface-enhanced Raman scattering (SERS) has long been recognized as an optical method that can provide fingerprint-like chemical information at the single-molecule level. Here, the advanced one-dimensional convolutional neural network (1D-CNN) approach was applied to accurately identify the SERS spectral signature of industrial wastewaters for source tracing. A total of 66,000 SERS spectra were acquired from wastewaters of 22 factories across 10 industrial categories at three excitation wavelengths after data augmentation. The dataset was used to train a 1D-CNN model consisting of three convolutional layers to achieve adequate feature extraction of SERS spectra. As a proof-of-concept, multimixed wastewater samples were used to simulate practical pollution scenarios and evaluate the application potential of the model. The SERS-1D-CNN platform can identify the amount and factory information of wastewaters in multimixed samples, which achieves a recognition accuracy rate of 97.33%. The results suggest that even in a complex and unknown water environment, the 1D-CNN model can accurately identify industrial wastewaters in precollected datasets, exhibiting excellent potential in pollution source tracing.

1. Introduction

Dissolved organic matter (DOM), a complex mixture of many organic compounds, has been intensely investigated for pollution source tracing (Ritson et al., 2019; Xu et al., 2021; Yang et al., 2015). The chemical composition of DOM can provide valuable fingerprint information for identifying different types of industrial wastewaters, possibly present in watersheds (Stedmon et al., 2003). In recent decades, ultraviolet visible absorption spectroscopy (UV–Vis) and fluorescence spectroscopy have been used to monitor DOM and distinguish the sources in natural environments and engineered systems (Birdwell and Engel, 2010; Henderson et al., 2009; Ishii and Boyer, 2012). However, both UV–Vis and fluorescence excitation-emission matrix methods can only be applied to DOM with specific chemical structures (chromophore or fluorophore), which are difficult to employ for the comprehensive characterization of DOM (Carstea et al., 2016). In addition, it has been reported that in DOM studies, fluorescence spectroscopy suffers greatly from weak signals and strong overlap between different spectral bands (Hu et al., 2020).

Surface-enhanced Raman scattering (SERS) has attracted

considerable attention due to its unique ability to collect abundant intrinsic fingerprint information and implement noninvasive data acquisition (Kneipp et al., 1997; Nie and Emory, 1997). The SERS effect relies on the excitation of localized surface plasmon resonances present on certain metallic nanostructures and additional charge-transfer contributions (Huang et al., 2020). The increase in Raman scattering efficiency enabled by SERS allows even single-molecule detection capability. In the last decade, advances in SERS substrates have encouraged the application of SERS to various systems of increasing complexity, such as biofluids (Bonifacio et al., 2015), biofilms (Ivleva et al., 2010), bacteria (Yang et al., 2016), living cells (Nam et al., 2019) and atmospheric aerosols (Feng et al., 2019). A label-free approach has been applied in most detection schemes, that is, all molecules present in the sample were allowed to interact with the surface of SERS substrates. This approach yields highly informative SERS spectra with very narrow and highly resolved bands that can simultaneously characterize multiple analytes. Moreover, the multiwavelength approach can capture different chemical functionalities in complex application scenarios (Nikitin et al., 2012). The vibrational features of analytes are associated with the

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excitation wavelength (Alvarez-Puebla, 2012), and two factors could be considered responsible for this behavior. The first is surface-enhanced resonance Raman scattering (SERS) (Fan et al., 2020; Lu et al., 2020), which will not only increase the signal intensity (up to 2 orders of magnitude compared with no resonance effect) but also change the spectral profile. Once the laser wavelength falls within the absorption range of the chemicals (and even the chromophore moieties), the signal from the specific chemicals and/or chemical moieties will dominate the observed spectrum. The second is the charge transfer effect (Sun et al., 2020), which will lead to a similar outcome. Chemicals directly adsorbed on the SERS substrate form complexes and have an absorption range that overlaps with the excitation laser wavelength. Thus, different spectral profiles are expected in multiple excitation wavelength conditions for the various chemical components in complex samples. Therefore, SERS is a potentially powerful technique to portray signatures of wastewaters for pollution source tracing in intricate watershed systems. However, due to the lack of advanced chemometrics methods to effectively extract spectral information from a large amount of complicated SERS spectra, previous studies of SERS in wastewater applications have mainly focused on detecting a single component, such as nitrate (Gajaraj et al., 2013), caffeine (Hu et al., 2018), methyl orange (Liu et al., 2020), p-nitroaniline (Lu et al., 2018) and antibiotics (Qu et al., 2016).

To address the issue of complicated spectral interpretation, mathematical and computational data processing based on a massive amount of SERS spectra was proposed (Lussier et al., 2020; Santos et al., 2017). In these regards, different methods can be employed, including the commonly used principal component analysis (PCA) (Shin et al., 2018; Sivashanmugan et al., 2019), partial least squares (Zhang et al., 2010), random forest (RF) (Amjad et al., 2018; Weng et al., 2018) and support vector machine (SVM) (Leong et al., 2021). Convolutional neural networks (CNNs) are an important branch of deep learning models, and exhibit some particular merits, including the requirement of little prior knowledge or design of explicit features and a strong capability to capture inner structures (Li et al., 2021). Unlike traditional machine learning algorithms, CNN discards the handcrafted features and directly obtains the mapping relation between the raw spectra and corresponding label information by the end-to-end learning process (Ding et al., 2021; Weng et al., 2020).

In this study, SERS spectra were used to characterize the spectral signatures for enhanced multiplex profiling of wastewaters, while the 1D-CNN model was built to implement wastewater identification and source tracing. A 1D-CNN model consisting of three convolutional layers was developed to extract the SERS spectral features of wastewaters, and its performance was compared with that of the RF and SVM approaches. Furthermore, actual pollution scenarios were simulated to explore the application potential of the wastewater tracing model in complicated and unknown water environments. As a proof-of-concept, the diurnal and seasonal variation of the wastewater sample from each industrial factory were not considered in this work.

2. Materials and methods

2.1. Materials and instruments

Silver nitrate (99%) and sodium citrate (99%) were purchased from Sigma-Aldrich (Shanghai China). Silver nanoparticles (Ag NPs) were prepared according to our previous report (Gong et al., 2014). A transmission electron microscope (TEM) (JSM 2100F, Japan) and a UV-Vis spectrometer (MAYA2000Pro, Ocean Optics, USA) were used to characterize the nanomaterials. Small-angle X-ray scattering (SAXS) (Empyrean, Netherlands) was performed to demonstrate the size distribution of the Ag NPs (Analysis software: Easy SAXS V. 2.2.0.1158; Analysis template: Template Type1&2 No. Fit. Saxsprt). The specific characterizations and descriptions can be found in Figure S1. Moreover, SERS measurements were performed on an XploRA PLUS Raman system

(HORIBA Scientific, Tokyo, Japan).

2.2. Wastewater samples

The wastewaters were sampled provincewide in Sichuan province from 22 factories belonging to 10 industrial categories, including resin and plastic (R), papermaking (Pa), food (F), petrochemistry (Pe), textile dyeing (T), furniture (Fu), car manufacture (Ca), pharmaceutical (Ph), domestic sewage (D) and steel (S). The effluent samples were taken from the final effluent outlet of each factory. The water quality parameters of the effluents, including pH, suspended solids (SS), biological oxygen demand (BOD), ammonia nitrogen ($\text{NH}_3\text{-N}$) and chemical oxygen demand (COD_{Cr}), were determined according to the national water standard methods in China. The analysis method and results are shown in Table S1.

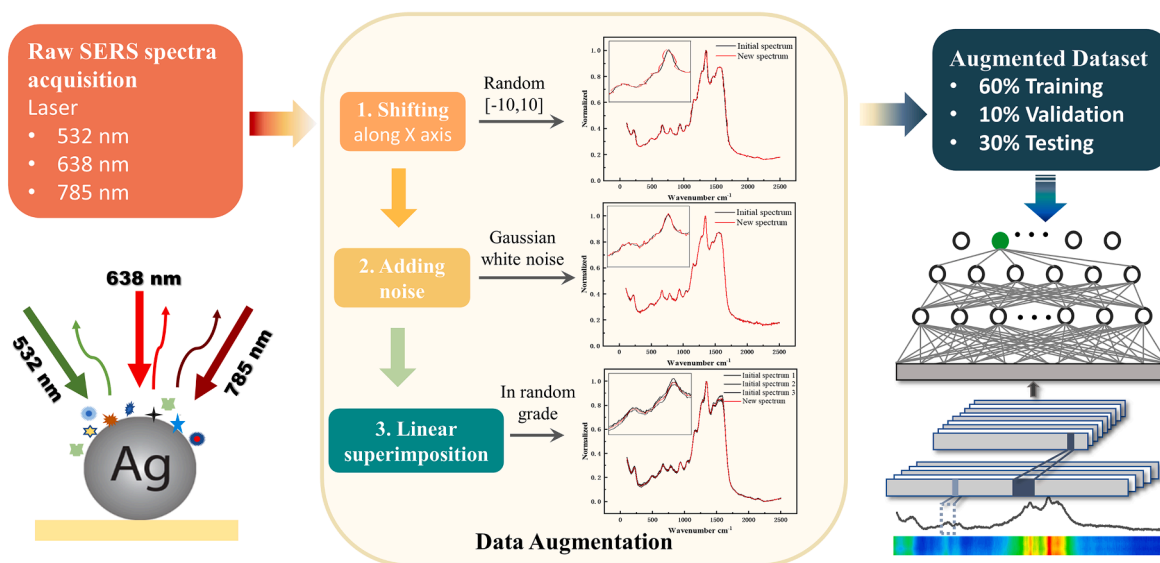
2.3. Data acquisition and augmentation

The data acquisition and augmentation framework are shown in [Schema 1](#). All wastewater samples were not preprocessed before analysis to retain more raw features. The Ag NP suspension and the wastewater samples were mixed in equal proportions, and 400 μL of the mixed solution was placed in a quartz cuvette (40 mm $\times$ 10 mm $\times$ 2 mm) for SERS analysis. Three excitation wavelengths (532, 638 and 785 nm) combined with a 50 $\times$ objective (numerical aperture (NA) = 0.60) were used and the laser power reaching the cuvette was controlled at approximately 10 mW. Considering the uneven distribution of the SERS effect (Fan et al., 2020) and various active chemical functionalities of DOM on Ag NPs, 29 individual SERS spectra of each wastewater sample were collected at each laser condition. As a result, a total of 1914 original SERS spectra were obtained. PCA was implemented to explore the superiority of SERS spectra in information capture of wastewaters and the specific discussions are shown in the supplementary information.

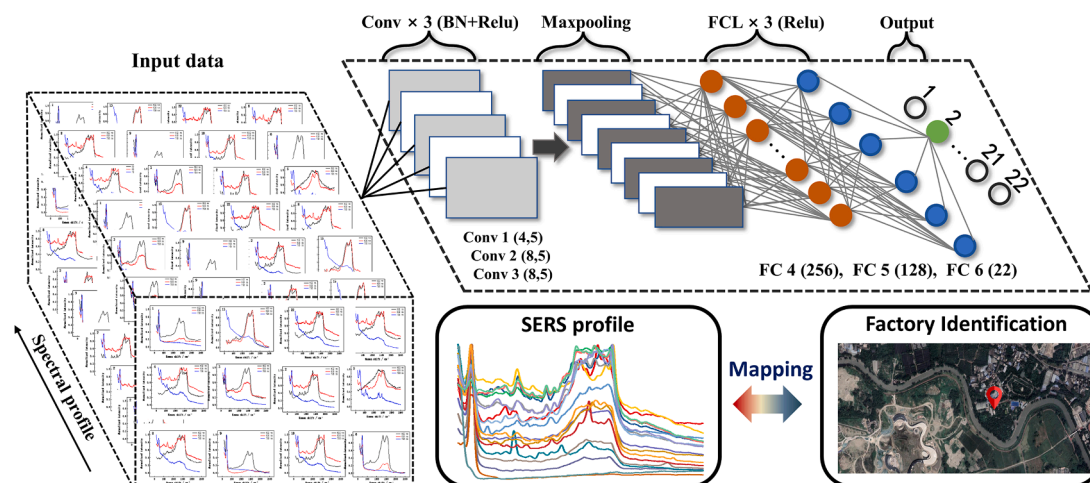
Data augmentation was employed in the 1D-CNN model training section, which can be regarded as injecting prior knowledge about invariant attributes of the data through some transformations, which can solve the problem of insufficient data, improve the robustness and accuracy of neural networks and avoid overfitting in network learning (Yang et al., 2019). Three strategies were adopted to realize the augmentation operation in this work, including spectral shift, noise injection and linear superposition. (1) The original spectra were shifted at random left or right along the horizontal axis in the range of 0–10 cm^{-1} to generate random spectra. As a result, every wastewater SERS spectrum dataset consisting of 29 original spectra was expanded to 50 spectra at each excitation wavelength. (2) By adding random Gaussian white noise with unequal intensities to the original and shifted SERS spectra, the number of SERS spectra for each wastewater was expanded to 200. (3) Linear superposition was performed using random scale coefficients with a total of 1 to expand the number of SERS spectrum datasets to 1000 for each wastewater sample. After the three data augmentation operations, 22,000 spectra were acquired at each laser wavelength.

2.4. Convolutional neural network

The classical network structure AlexNet is employed to construct the 1D-CNN model in this work (Krizhevsky et al., 2012; Turaga et al., 2010), which was carried out on PyTorch 1.6.0. As shown in [Schema 2](#), the 1D-CNN model consists of an input layer, three convolutional layers, three fully connected layers, and an output layer. Each convolutional layer is followed by a batch normalization (BN) layer, a rectified linear unit (ReLU) layer and a max pooling layer. Multiple convolutional layers were used to extract high-level abstractions of the input automatically, while fully connected layers were employed to transform the multilayered features into the required output form.



Schema 1. The data acquisition and augmentation framework.



Schema 2. Architecture of the one-dimensional convolutional neural network. FC: fully connected layer; Conv: convolutional layers; BN: batch normalization layer; ReLU: rectified linear unit layer.

The identification process was performed by the 1D-CNN model with the following procedures: (1) SERS spectrum data are input into the network. (2) Hierarchical representations (features) of the inputs are learned by multiple convolutional layers. (3) The representations are concatenated as a feature vector to be fed into the fully connected layer to generate a weighted sum. (4) A class label, as the network output, is predicted with the class likelihood values from the softmax (an activation function that would yield a likelihood value for each class) layer using the weighted sum. The optimizer based on the chain rule is used to improve the performance of the 1D-CNN model with backward propagation, which makes the model converge faster and more reliably. The overall training process is shown in Schema S2. Once the training step is completed, the classification pattern information is implicitly stored in the network. Moreover, the cross-entropy loss function is employed to estimate and measure the performance of the 1D-CNN model (Wang and Kumar, 2019), which is further introduced in the supplementary

information.

In addition, the acquired 22,000 spectra at each laser wavelength after data augmentation were split randomly for training, validating, and testing the 1D-CNN. Sixty percent of the data are used as a training dataset for the model to learn the characteristics of SERS spectra. Ten percent of the data are used as the validation dataset, and the remaining 30% of the data are used as the testing dataset. More details of the algorithm design of the 1D-CNN are described in the supplementary information and Section 3.3.

2.5. RF and SVM

The RF and SVM were chosen as representatives of the classical algorithm to compare with the 1D-CNN model in this work. The operation of SVM is based on the LIBSVM toolkit. The Gaussian radial basis function is used as the kernel function, and the best penalty parameter C

and kernel parameter g are selected through cross-validation. Both RF and SVM were implemented in MATLAB R2020a. Similarly, the spectral data is also divided into training, validation and testing dataset in a ratio of 6:1:3 and the dataset used for RF and SVM was the same as the training, validation, and test datasets used for 1D-CNN.

2.6. Source tracing model

In the actual scenarios of tracing pollution sources, the analyte is usually a mixture of many discharge sources. Therefore, it is necessary to identify the discharge sources in complex water samples filled with extensive unknown compositions and verify which discharge source is the primary factor in contamination accidents. In this Section, the source tracing scheme was adapted from the wastewater identification model constructed in [Section 2.4](#), while the SERS spectral characteristics of the 22 industrial wastewaters in multiple convolutional layers were retained. The sigmoid function was used to replace the softmax function in the output layer of the wastewater identification model, forming the wastewater tracing model.

The testing dataset in this section comes from the mixed wastewater samples among the 22 industrial wastewaters, including 21 two-mixed water samples, 13 three-mixed water samples, 9 four-mixed water samples, 4 five-mixed water samples and 3 six-mixed water samples. All mixed water samples were blended in equal proportions and the details of the mixed arrays are shown in Table S2. There were 50 mixed water samples in total, and 30 SERS spectra of each mixed water sample were collected at three excitation wavelengths. Therefore, 1500 labeled SERS spectra at each excitation wavelength were recorded.

3. Results and discussion

3.1. Water characteristics of industrial effluents

Fig. 1 shows the visualized comparison of five crucial water quality parameters that are related to wastewater quality, and the specific information can be seen in Table S1. The water quality parameters of wastewaters from the same industrial category do not show high

similarities. Generally, SS, BOD and COD are higher in wastewaters from food, steel, petrochemical and car manufacturing facilities. The average value in $\text{NH}_3\text{-N}$ of all wastewater samples was 33.3 mg/L. The wastewater from car manufacturing (No. 12) shows the lowest value (1.49 mg/L), which is well below the average. However, there is no statistically significant difference in SS, BOD, $\text{NH}_3\text{-N}$ and COD_{Cr} among the 22 industrial wastewaters based on variance analysis (Table S3), which may be due to the high variation in the water quality parameter for each industrial category. The significant difference in pH is attributed to the pharmaceutical industry wastewater (No. 15, $\text{pH} = 2.74$). Common water quality parameters can be used to evaluate the impact of wastewater on the environment, but it is difficult to describe the special features of wastewater.

3.2. SERS spectral signature of industrial effluents

The averaged SERS spectra of some effluents at three excitation wavelengths and the corresponding PCA scatter plot are shown in Fig. 2. The spectra of the rest of the wastewaters can be seen in Table S4. All spectra were normalized by linearly rescaling the intensity values to the range [0,1] to bring the peaks to comparable heights, allowing easier comparison among SERS spectra. A majority of spectral profiles at different excitation wavelengths show a significant variation from the PCA results. The spectra of samples 2 and 10 show a high similarity at 532 nm, and even some subtle spectral features, including shoulder peaks at 1061, 1282, and 1460 cm^{-1} are fairly uniform. However, their unique characteristic peaks at 1340 and 1616 cm^{-1} can be captured at 638 nm. Thus, the multiwavelength approach can provide differentiated SERS spectral characteristics of wastewaters, which can be reflected in the spectral datasets.

As shown in Fig. 2 and Table S4, a more intense fluorescence background is observed at 532 and 638 nm. It is reasonable that the fluorescence interference encountered for many compounds is caused by visible excitation, which can be mediated by using long-wavelength (785 nm) excitation (Larkin, 2011). In addition, the SERS spectra provide much composition information on the wastewaters. For example, there are typically broad bands with peaks at approximately 1080–1085

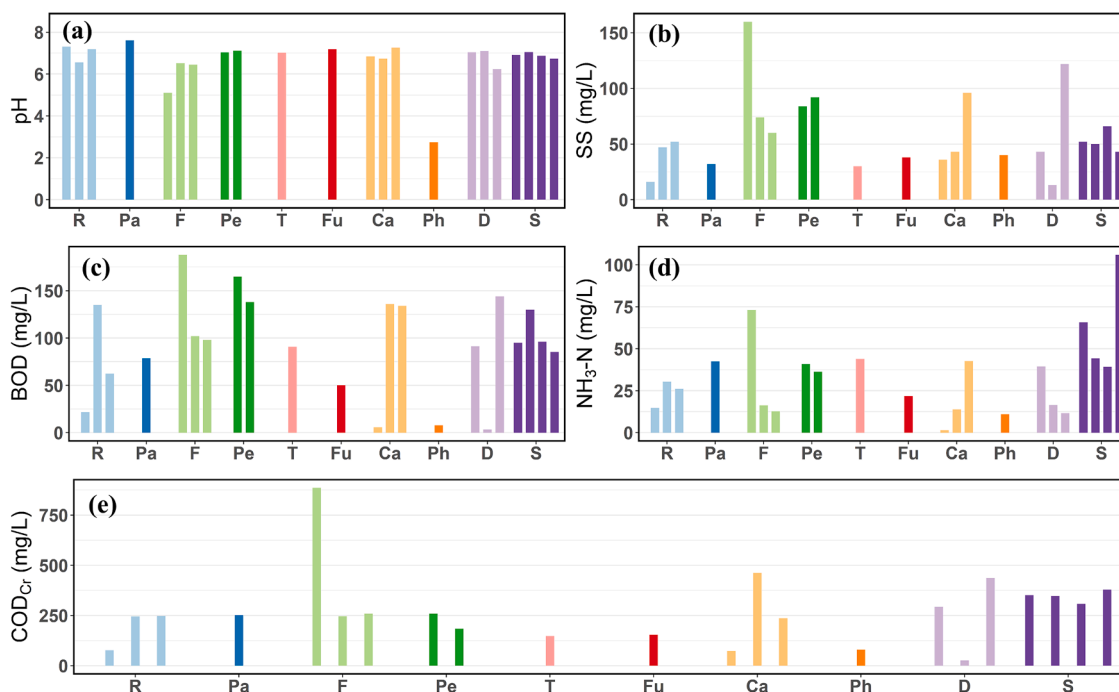


Fig. 1. The water quality parameters of the industrial wastewaters: (a) pH; (b) SS; (c) BOD; (d) $\text{NH}_3\text{-N}$; (e) COD_{Cr} . R: Resin and plastic; Pa: Papermaking; F: Food; Pe: Petrochemical; T: Textile dyeing; Fu: Furniture; Ca: Car manufacture; Ph: Pharmaceutical; D: Domestic sewage; S: Steel.

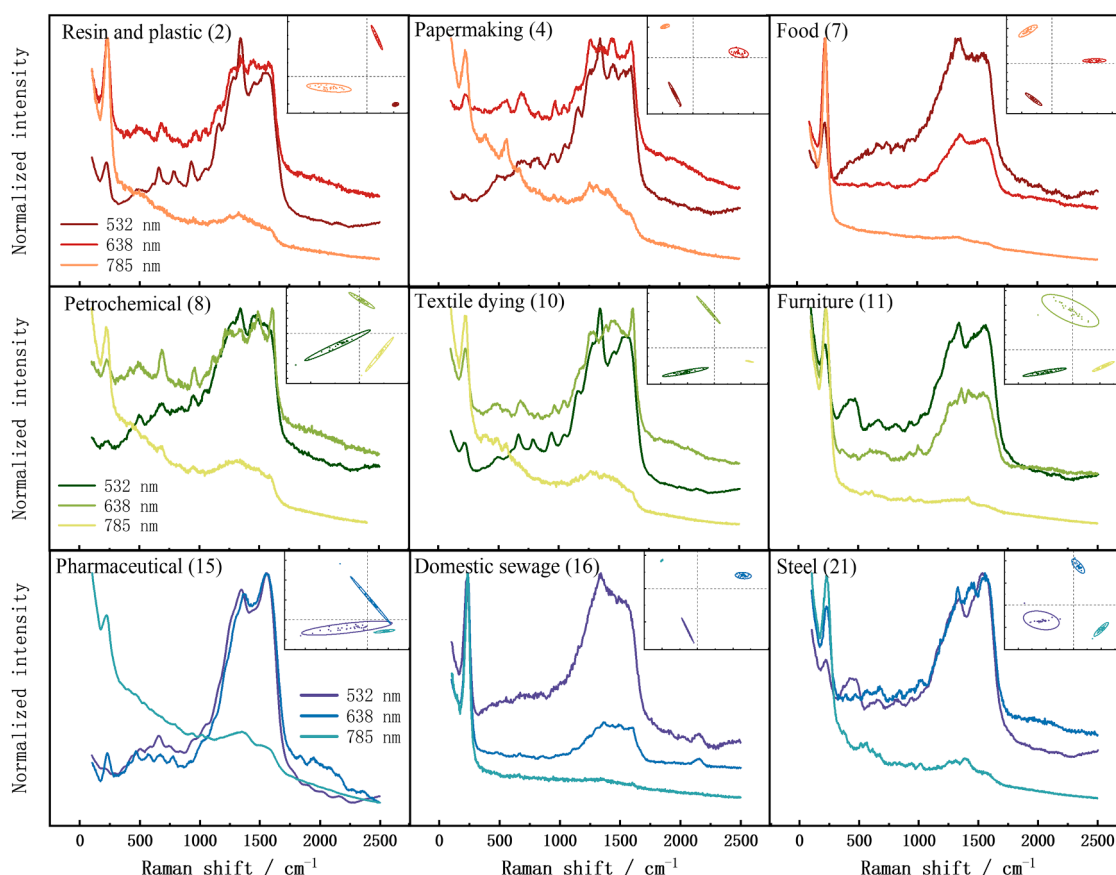


Fig. 2. Normalized SERS spectra of wastewaters from different industrial categories, including resin and plastic (2), papermaking (4), food (7), petrochemical (8), textile dyeing (10), furniture (11), pharmaceutical (15), domestic sewage (16) and steel (21) at three excitation wavelengths. (inset) PCA scatter plots of the corresponding SERS spectra, and the numbers in parentheses indicate the wastewater sample number.

and 1135–1295 cm^{-1} due to the OH stretching of hydroxyl/carboxyl groups and the NH stretching of amides, while the methyl/methylene CH stretching bands typically appear at 305–480 and 1000–1020 cm^{-1} . The band assigned to the aromatic ring can appear in the SERS spectrum in the region of 1320–1340 cm^{-1} . The peak at 530–580 cm^{-1} due to polysaccharide CO stretching was present, with a secondary peak at 1090–1095 cm^{-1} . More functional groups and compounds can be inferred by the peak position of the SERS spectra combined with peak assignment in Table S5. However, specific composition analysis based on peak assignment is inaccessible.

In addition, the superiority of SERS spectra in information capture of wastewaters was explored by PCA, and the specific discussions are shown in the supplementary information. From the results, we see that the SERS spectra can indeed reflect subtle changes in wastewater composition, exhibiting potential application as spectral fingerprints in discriminating different wastewaters and tracing pollution sources. However, the PCA results cannot give specific and valuable identification information of wastewaters, so the more advanced CNN algorithm was introduced.

3.3. 1D-CNN development for wastewater identification

The classical AlexNet contains eight learned layers: the first five are convolutional and the remaining three are fully connected layers (Krizhevsky et al., 2017). Herein, the number of convolutional layers is adjusted to explore the optimal network depth accommodating the SERS spectral data, while the three fully connected layers are reserved. From Figure S5, we see that the recognition accuracy reaches the maximum when the number of convolutional layers is 3, which is also the critical

value of network depth for the spectral data. Meanwhile, the classification process also has a short running time. Hence, the 1D-CNN network with three convolutional layers and three fully connected layers was determined.

The loss curves and the accuracy curves are shown in Fig. 3(a) and (b), respectively. The optimal hyperparameters were tuned during classifier construction and are listed in Table S7. The initialization of the 1D-CNN parameters is random, which is completely different from the final model. Thus, a significant error was observed at the beginning. As the number of iterations increased, the model parameters quickly updated during the initial training. Therefore, as depicted in Fig. 3(a), the prediction loss begins at a high level and drops exponentially over relatively short periods. At the same time, the training accuracy and validation accuracy demonstrated in Fig. 3(b) increase exponentially and reach a plateau from the 20th epoch.

From the accuracy rate of different algorithms at three laser wavelengths shown in Table 1, we see that the spectral features at 638 nm are more abundant than those at other wavelengths, so that the RF, SVM and 1D-CNN algorithms have the highest accuracy rates at 638 nm, 89.65%, 93.45% and 100% respectively. The specific performance breakdown for individual classification can be seen in the confusion matrix in Figure S6. In contrast, the 1D-CNN has excellent identification performance (the accuracy rates all reach 100% at three laser wavelengths), which shows a powerful feature mining, analysis and recognition capability.

Traditional supervised learning algorithms such as RF and SVM require prespecified handcrafted features that hinge on prior domain knowledge of extensive experience. In contrast, the 1D-CNN has direct access to the raw data and learns the features automatically during the

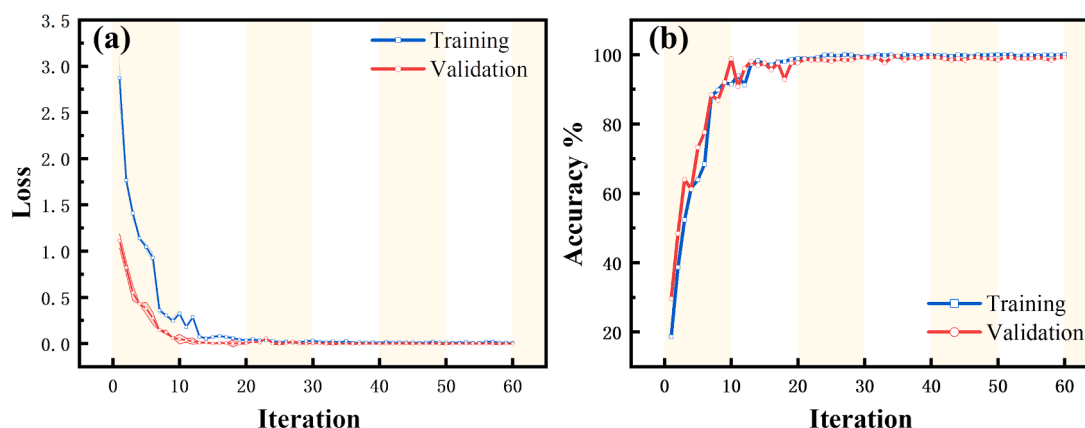


Fig. 3. (a) Loss curves and (b) accuracy curves of the 1D-CNN model.

Table 1

Accuracy rates of different algorithms at three laser wavelengths^a.

Method	Accuracy rate% 532nm	638nm	785nm
1D-CNN	100	100	100
RF	71.95	89.65	47.65
SVM	77.75	93.45	45.95

^a The average of 10 tests.

network training process. In this scenario, the 1D-CNN is free to use all levels of features and learn more semantic information from the given data so that it can better reflect the differences among the targets.

3.4. SERS-1D-CNN application in source tracing

In the actual scenarios of tracing pollution sources, the overall accuracy rate, category accuracy rate and incomplete accuracy rate were used to evaluate the performance of the wastewater tracing model and are shown in Table 2.

The overall accuracy rate is lower than the category accuracy rate at the three excitation wavelengths, which means that there is some incorrect assignment of the specific factory information. This is reasonable since some spectra exhibit high similarity even from different wastewater samples, such as samples 2 and 10. Moreover, the overall accuracy rate of the model is less than the incomplete accuracy rate, showing that the model has a lack of traceability for some wastewater samples. The incomplete prediction events usually occurred on the four, five and six-mixed water samples, showing that the cover-up effect in the complicated mix scenario will increase the difficulty of source identification. However, even at the relatively poor performance at 785 nm, the overall accuracy rate remains above 94%, indicating that the wastewater tracing model has excellent potential in source tracing.

The 1D-CNN model in this work, including the wastewater identification model and wastewater tracing model, can be applied to other

Table 2

Accuracy rate of the wastewater tracing model at three excitation wavelengths.

Laser wavelength	Overall accuracy rate (%) ^a	Category accuracy rate (%) ^b	Incomplete accuracy rate (%) ^c
532 nm	96.13	96.67	97.07
638 nm	97.33	97.93	98.40
785 nm	94.67	95.13	95.33

^a Both the amount of mixed wastewaters and factory information are correct.

^b The number of mixed wastewaters is correct.

^c The number of mixed wastewaters is less than or equal to the actual value, and the factory information is correct. The results come from the average of 10 runs.

identification and tracking scenarios with minimal modification such as bacteria identification and air pollution tracing, or other spectroscopic techniques such as infrared and mass spectrometry. It should be noted that this work is a proof-of-concept. To achieve the application of the SERS-1D-CNN platform in water pollution incidents as fine-grained as those regulatory methods, larger datasets covering more facilities in the same factory, more spectral profiles of the individual sample, more factories from different industrial categories and continuous all-year sampling of watershed water samples will be needed. Although collecting such datasets is beyond an academic scope, requiring highly automated sample preparation and data acquisition processes, there is great promise for high-efficiency water pollution source tracing.

4. Conclusions

This study preliminarily explored the application potential of SERS spectra combined with 1D-CNN in identifying industrial wastewaters and distinguishing water pollution sources. As an automatic feature extractor, the 1D-CNN model can distinguish the effluents with a 100% recognition accuracy rate, which shows a better classification performance than the traditional models/algorithms (RF and SVM) based on handcrafted features. Furthermore, multimixed wastewater samples simulating practical pollution source tracing scenarios were implemented to verify the application potential of the model. The SERS-1D-CNN platform can identify the amount and factory information of wastewaters in multimixed samples, which achieves a recognition accuracy rate of 97.33%. The results suggest that the SERS-1D-CNN platform has significant application potential in pollution source tracing.

CRedit authorship contribution statement

Yuting Huang: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft. **Bingxue Yuan:** Software, Investigation, Formal analysis. **Xueqing Wang:** Software, Investigation, Formal analysis. **Yongsheng Dai:** Methodology, Software. **Dongmei Wang:** Project administration. **Zhengjun Gong:** Project administration. **Junmin Chen:** Resources. **Li Shen:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing. **Meikun Fan:** Conceptualization, Supervision, Funding acquisition, Writing – review & editing. **Zhilin Li:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2023.119662.

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